

STATISTICAL SUMMARY OF GENOTYPE TRENDS

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1. OVERALL PATTERNS

- Total Records: 8,890
- Time Period: 2015-2024 (10 years)
- Number of Years with Data: 10

2. CLINICAL CATEGORY TRENDS

Normal (AA) Pattern:

- Annual Rate of Change: 0.3086 percentage points per year
- Total Change (2015-2024): +2.78 percentage points
- Statistical Significance: not significant ($p=0.2510$)

Carrier (AS/AC) Pattern:

- Annual Rate of Change: -0.2780 percentage points per year
- Total Change (2015-2024): -2.50 percentage points
- Statistical Significance: not significant ($p=0.3063$)

Disease (SS/SC/CC) Pattern:

- Annual Rate of Change: -0.0301 percentage points per year
- Total Change (2015-2024): -0.27 percentage points
- Statistical Significance: not significant ($p=0.4568$)

3. KEY FINDINGS

- Stable Distribution: The overall genotype distribution has remained relatively stable over the 10-year period.
- Dominant Genotype: AA genotype consistently accounts for approximately 75% of the population.
- Carrier Prevalence: AS/AC carriers maintain around 23-24% of the population.
- Disease Burden: SS/SC/CC disease genotypes remain low at about 1-1.5%.

4. STATISTICAL INTERPRETATION

- The trends show minimal variation over time.
- No significant shifts in genetic distribution patterns.
- Consistent with Hardy-Weinberg equilibrium in a stable population.
- Gender ratios remain balanced across the study period.